20261019 Quiz

for (MATH210) Applied Complex Variables

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Exercise 47.3. Show that if C is the boundary of the triangle with vertices at the points 0, $3i$, and -4, oriented in the counterclockwise direction (see Fig. 50), then

$$\left|\int_C (e^z - \overline{z})\, \mathrm{d}z\right| \leq 60.$$

Suggestion. Note that $|e^z - \overline{z}| \leq e^x + \sqrt{x^2 + y^2}$ when $z = x + iy$.

Solution. Every point $z = x + iy$ in the closed triangle satisfies $x \leq 0$ and $|z| \leq 4$. Indeed, each point is a convex combination of 0, $3i$, and -4, so the triangle inequality gives $|z| \leq 4$. Hence, on C,

$$|e^z - \overline{z}| \leq |e^z| + |z| = e^x + |z| \leq 1 + 4 = 5.$$

The three sides have lengths 3, 4, and 5, respectively; thus $L(C) = 12$. The ML inequality now yields

$$\boxed{\left|\int_C (e^z - \overline{z})\, \mathrm{d}z\right| \leq 5 \cdot 12 = 60.}$$

□

Exercise 47.4. Let C_R denote the upper half of the circle $|z| = R$ $(R > 2)$, taken in the counterclockwise direction. Show that

$$\left|\int_{C_R} \frac{2z^2 - 1}{z^4 + 5z^2 + 4}\, \mathrm{d}z\right| \leq \frac{\pi R(2R^2 + 1)}{(R^2 - 1)(R^2 - 4)}.$$

Then, by dividing the numerator and denominator on the right here by R^4, show that the value of the integral tends to zero as R tends to infinity. (Compare with Example 2 in Sec. 47.)

Solution. Since $z^4 + 5z^2 + 4 = (z^2 + 1)(z^2 + 4)$, for $|z| = R > 2$ we have

$$|2z^2 - 1| \leq 2R^2 + 1, \qquad |z^2 + 1| \geq R^2 - 1, \qquad |z^2 + 4| \geq R^2 - 4.$$

The length of C_R is πR. Therefore the ML inequality gives

$$\boxed{\left|\int_{C_R} \frac{2z^2 - 1}{z^4 + 5z^2 + 4}\, \mathrm{d}z\right| \leq \frac{\pi R(2R^2 + 1)}{(R^2 - 1)(R^2 - 4)}.}$$

Writing the upper bound as

$$\frac{\pi(2/R + 1/R^3)}{(1 - 1/R^2)(1 - 4/R^2)} \longrightarrow 0 \qquad (R \to \infty),$$

we conclude by the squeeze theorem that the contour integral tends to 0.

□

Exercise 47.5. Let C_R be the circle $|z| = R$ $(R > 1)$, described in the counterclockwise direction. Show that

$$\left| \int_{C_R} \frac{\operatorname{Log} z}{z^2} \, \mathrm{d}z \right| < 2\pi \left(\frac{\pi + \ln R}{R} \right),$$

and then use l'Hospital's rule to show that the value of this integral tends to zero as R tends to infinity.

Solution. On C_R, write $\operatorname{Log} z = \ln R + i \operatorname{Arg} z$, where $-\pi < \operatorname{Arg} z \leq \pi$. Since $\ln R > 0$,

$$|\operatorname{Log} z| \leq \sqrt{(\ln R)^2 + \pi^2} < \ln R + \pi.$$

Thus the modulus of the integrand is strictly less than $(\ln R + \pi)/R^2$. Since $L(C_R) = 2\pi R$, the ML inequality gives

$$\boxed{\left| \int_{C_R} \frac{\operatorname{Log} z}{z^2} \, \mathrm{d}z \right| < 2\pi \frac{\pi + \ln R}{R}.}$$

Here the principal logarithm has a jump at $z = -R$; the integral is understood piecewise along the parametrized circle, with the one-sided limiting values at that point. This does not affect the estimate. Finally,

$$\lim_{R \to \infty} \frac{\ln R}{R} \overset{\text{L'Hôpital}}{=} \lim_{R \to \infty} \frac{1/R}{1} = 0,$$

so the integral tends to 0.

□

Exercise 47.6. Let C_ρ denote a circle $|z| = \rho$ $(0 < \rho < 1)$, oriented in the counterclockwise direction, and suppose that $f(z)$ is analytic in the disk $|z| \leq 1$. Show that if $z^{-1/2}$ represents any particular branch of that power of z, then there is a nonnegative constant M, independent of ρ, such that

$$\left| \int_{C_\rho} z^{-1/2} f(z) \, \mathrm{d}z \right| \leq 2\pi M \sqrt{\rho}.$$

Thus show that the value of the integral here approaches 0 as ρ tends to 0.

Suggestion. Note that since $f(z)$ is analytic, and therefore continuous, throughout the disk $|z| \leq 1$, it is bounded there (Sec. 18).

Solution. By continuity on the closed unit disk, there is a constant

$$M = \max_{|z| \leq 1} |f(z)| \geq 0.$$

For $|z| = \rho$, every value of the power $z^{-1/2}$ has modulus $\rho^{-1/2}$. Consequently,

$$|z^{-1/2} f(z)| \leq \frac{M}{\sqrt{\rho}}.$$

As $L(C_\rho) = 2\pi\rho$, the ML inequality yields

$$\boxed{\left| \int_{C_\rho} z^{-1/2} f(z) \, \mathrm{d}z \right| \leq \frac{M}{\sqrt{\rho}} (2\pi\rho) = 2\pi M \sqrt{\rho}.}$$

A fixed analytic branch of $z^{-1/2}$ cannot be defined on a neighborhood of the entire circle; if its cut meets C_ρ, the integral is interpreted on the resulting arcs using one-sided values. The bound is valid for every such choice. Since M does not depend on ρ, the integral tends to 0 as $\rho \to 0^+$.

□

Exercise 49.2. By finding an antiderivative, evaluate each of these integrals, where the path is any contour between the indicated limits of integration:

(a) $\displaystyle\int_0^{1+i} z^2 \, dz$;

(b) $\displaystyle\int_0^{\pi+2i} \cos\left(\frac{z}{2}\right) dz$;

(c) $\displaystyle\int_1^3 (z-2)^3 \, dz$.

Solution. Each integrand has an entire antiderivative, so the integral depends only on its endpoints.

(a) With $F(z) = z^3/3$,

$$\int_0^{1+i} z^2 \, dz = \frac{(1+i)^3}{3} = \boxed{\frac{2}{3}(-1+i)}.$$

(b) With $F(z) = 2\sin(z/2)$,

$$\int_0^{\pi+2i} \cos\left(\frac{z}{2}\right) dz = 2\sin\left(\frac{\pi}{2}+i\right) = 2\cosh 1 = \boxed{e+e^{-1}}.$$

(c) With $F(z) = (z-2)^4/4$,

$$\int_1^3 (z-2)^3 \, dz = \frac{(3-2)^4 - (1-2)^4}{4} = \boxed{0}.$$

□

Exercise 49.3. Use the theorem in Sec. 48 to show that

$$\int_{C_0} (z-z_0)^{n-1} \, dz = 0 \qquad (n = \pm 1, \pm 2, \ldots)$$

when C_0 is any closed contour which does not pass through the point z_0. (Compare with Exercise 13, Sec. 46.)

Solution. For every nonzero integer n, the function

$$F(z) = \frac{(z-z_0)^n}{n}$$

is single-valued and analytic on $\mathbb{C} \setminus \{z_0\}$, with

$$F'(z) = (z-z_0)^{n-1}.$$

Since C_0 lies in this domain, the antiderivative theorem of Sec. 48 applies even if C_0 surrounds z_0. The initial and terminal points coincide, and hence

$$\boxed{\int_{C_0} (z-z_0)^{n-1} \, dz = 0 \qquad (n \in \mathbb{Z} \setminus \{0\}).}$$

□

Exercise 49.5. Show that

$$\int_{-1}^{1} z^i \, \mathrm{d}z = \frac{1 + e^{-\pi}}{2}(1 - i),$$

where the integrand denotes the principal branch

$$z^i = \exp(i \operatorname{Log} z) \qquad (|z| > 0, \ -\pi < \operatorname{Arg} z < \pi)$$

of z^i and where the path of integration is any contour from $z = -1$ to $z = 1$ that, except for its end points, lies above the real axis. (Compare with Exercise 6, Sec. 46.)

Suggestion. Use an antiderivative of the branch

$$z^i = \exp(i \log z) \qquad \left(|z| > 0, \ -\frac{\pi}{2} < \arg z < \frac{3\pi}{2}\right)$$

of the same power function.

Solution. Choose the analytic logarithm L with

$$L(z) = \ln |z| + i\theta(z), \qquad -\frac{\pi}{2} < \theta(z) < \frac{3\pi}{2}.$$

Its domain contains both endpoints and the entire given contour, since the latter lies in the upper half-plane except at its endpoints. Moreover, L agrees with the principal logarithm at points strictly above the real axis. Put

$$f(z) = e^{iL(z)}, \qquad F(z) = \frac{e^{(1+i)L(z)}}{1 + i}.$$

Since $L'(z) = 1/z$, differentiation gives

$$F'(z) = \frac{1 + i}{1 + i} \frac{e^{(1+i)L(z)}}{z} = e^{iL(z)} = f(z).$$

The endpoint values are $L(1) = 0$ and $L(-1) = i\pi$; hence

$$F(1) = \frac{1}{1 + i}, \qquad F(-1) = \frac{e^{(1+i)i\pi}}{1 + i} = -\frac{e^{-\pi}}{1 + i}.$$

Thus

$$\boxed{\int_{-1}^{1} z^i \, \mathrm{d}z = F(1) - F(-1) = \frac{1 + e^{-\pi}}{1 + i} = \frac{1 + e^{-\pi}}{2}(1 - i).}$$

Strictly speaking, the principal branch in the question is undefined at -1. The displayed integral uses its limit from the upper half-plane; the branch L supplies its continuous extension there. □

Exercise 53.2. Let C_1 denote the positively oriented boundary of the square whose sides lie along the lines $x = \pm 1$, $y = \pm 1$ and let C_2 be the positively oriented circle $|z| = 4$ (Fig. 65). With the aid of the corollary in Sec. 53, point out why

$$\int_{C_1} f(z) \, \mathrm{d}z = \int_{C_2} f(z) \, \mathrm{d}z$$

when

(a) $f(z) = \dfrac{1}{3z^2 + 1}$;

(b) $f(z) = \dfrac{z + 2}{\sin(z/2)}$;

(c) $f(z) = \dfrac{z}{1 - e^z}$.

Solution. The square is strictly inside the circle. By the corollary in Sec. 53, it suffices in each case to check that f is analytic on and throughout the closed region between C_1 and C_2.

(a) The only singularities occur at $z = \pm i/\sqrt{3}$. Both lie strictly inside C_1, since $1/\sqrt{3} < 1$.

(b) The zeros of $\sin(z/2)$ are $z = 2k\pi$, $k \in \mathbb{Z}$. The origin lies inside C_1, while $|2k\pi| \geq 2\pi > 4$ for $k \neq 0$.

(c) The zeros of $1 - e^z$ are $z = 2k\pi i$, $k \in \mathbb{Z}$. The apparent singularity at $z = 0$ is removable because

$$\lim_{z \to 0} \frac{z}{1 - e^z} = -1.$$

All other singularities lie outside C_2, since $|2k\pi i| \geq 2\pi > 4$ for $k \neq 0$.

Therefore the required analyticity condition holds in all three cases, and

$$\boxed{\int_{C_1} f(z)\ \mathrm{d}z = \int_{C_2} f(z)\ \mathrm{d}z.}$$

□

Exercise 53.3. If C_0 denotes a positively oriented circle $|z - z_0| = R$, then

$$\int_{C_0} (z - z_0)^{n-1}\ \mathrm{d}z = \begin{cases} 0, & n = \pm 1, \pm 2, \ldots, \\ 2\pi i, & n = 0, \end{cases}$$

according to Exercise 13, Sec. 46. Use that result and the corollary in Sec. 53 to show that if C is the boundary of the rectangle $0 \leq x \leq 3$, $0 \leq y \leq 2$, described in the positive sense, then

$$\int_C (z - 2 - i)^{n-1}\ \mathrm{d}z = \begin{cases} 0, & n = \pm 1, \pm 2, \ldots, \\ 2\pi i, & n = 0. \end{cases}$$

Solution. Put $z_0 = 2 + i$. This point is strictly inside the rectangle, so choose $0 < R < 1$ and let C_0 be the positively oriented circle $|z - z_0| = R$. Then C_0 lies strictly inside C. For every integer n, the function $(z - z_0)^{n-1}$ is analytic at every point between and on C_0 and C. The deformation corollary in Sec. 53 therefore gives

$$\int_C (z - z_0)^{n-1}\ \mathrm{d}z = \int_{C_0} (z - z_0)^{n-1}\ \mathrm{d}z.$$

Applying the supplied circle integral yields

$$\boxed{\int_C (z - 2 - i)^{n-1}\ \mathrm{d}z = \begin{cases} 0, & n \in \mathbb{Z} \setminus \{0\}, \\ 2\pi i, & n = 0. \end{cases}}$$

□

Exercise 53.4. Use the following method to derive the integration formula

$$\int_0^\infty e^{-x^2} \cos(2bx)\ \mathrm{d}x = \frac{\sqrt{\pi}}{2} e^{-b^2} \qquad (b > 0).$$

(a) Show that the sum of the integrals of e^{-z^2} along the lower and upper horizontal legs of the rectangular path in Fig. 66 (with vertices $-a$, a, $a + ib$, and $-a + ib$, in counterclockwise order) can be written

$$2\int_0^a e^{-x^2}\ \mathrm{d}x - 2e^{b^2} \int_0^a e^{-x^2} \cos(2bx)\ \mathrm{d}x,$$

and that the sum of the integrals along the vertical legs on the right and left can be written

$$ie^{-a^2}\int_0^b e^{y^2}e^{-i2ay}\,\mathrm{d}y - ie^{-a^2}\int_0^b e^{y^2}e^{i2ay}\,\mathrm{d}y.$$

Thus, with the aid of the Cauchy–Goursat theorem, show that

$$\int_0^a e^{-x^2}\cos(2bx)\,\mathrm{d}x = e^{-b^2}\int_0^a e^{-x^2}\,\mathrm{d}x + e^{-(a^2+b^2)}\int_0^b e^{y^2}\sin(2ay)\,\mathrm{d}y.$$

(b) By accepting the fact that

$$\int_0^\infty e^{-x^2}\,\mathrm{d}x = \frac{\sqrt{\pi}}{2}$$

and observing that

$$\left|\int_0^b e^{y^2}\sin(2ay)\,\mathrm{d}y\right| \le \int_0^b e^{y^2}\,\mathrm{d}y,$$

obtain the desired integration formula by letting a tend to infinity in the equation at the end of part (a).

Solution. Let C be the positively oriented rectangle with vertices $-a$, a, $a+ib$, and $-a+ib$, where $a, b > 0$.

(a) On the lower edge $z = x$ with $-a \le x \le a$, and on the upper edge $z = x + ib$ with $a \ge x \ge -a$. Hence the sum of the two horizontal integrals is

$$\begin{aligned}&\int_{-a}^a e^{-x^2}\,\mathrm{d}x - e^{b^2}\int_{-a}^a e^{-x^2}e^{-2ibx}\,\mathrm{d}x\\ &\qquad = 2\int_0^a e^{-x^2}\,\mathrm{d}x - 2e^{b^2}\int_0^a e^{-x^2}\cos(2bx)\,\mathrm{d}x,\end{aligned}$$

since the imaginary part of the second integral is odd in x. On the right edge $z = a + iy$ with $0 \le y \le b$, and on the left edge $z = -a + iy$ with $b \ge y \ge 0$. Their sum is

$$\begin{aligned}&ie^{-a^2}\int_0^b e^{y^2}e^{-2iay}\,\mathrm{d}y - ie^{-a^2}\int_0^b e^{y^2}e^{2iay}\,\mathrm{d}y\\ &\qquad = 2e^{-a^2}\int_0^b e^{y^2}\sin(2ay)\,\mathrm{d}y,\end{aligned}$$

using $e^{-it} - e^{it} = -2i\sin t$. As e^{-z^2} is entire, the Cauchy–Goursat theorem shows that the sum of all four integrals is zero. Dividing the resulting equality by $2e^{b^2}$ gives

$$\boxed{\int_0^a e^{-x^2}\cos(2bx)\,\mathrm{d}x = e^{-b^2}\int_0^a e^{-x^2}\,\mathrm{d}x + e^{-(a^2+b^2)}\int_0^b e^{y^2}\sin(2ay)\,\mathrm{d}y.}$$

(b) For fixed $b > 0$, the last term satisfies

$$\left|e^{-(a^2+b^2)}\int_0^b e^{y^2}\sin(2ay)\,\mathrm{d}y\right| \le e^{-(a^2+b^2)}\int_0^b e^{y^2}\,\mathrm{d}y \longrightarrow 0$$

as $a \to \infty$. Also, $e^{-x^2}|\cos(2bx)| \le e^{-x^2}$, so the integral on the left converges absolutely. Taking the limit in part (a) and using the given Gaussian integral, we obtain

$$\boxed{\int_0^\infty e^{-x^2}\cos(2bx)\,\mathrm{d}x = e^{-b^2}\frac{\sqrt{\pi}}{2}.}$$

□

Exercise 53.7. Show that if C is a positively oriented simple closed contour, then the area of the region enclosed by C can be written

$$\frac{1}{2i}\int_C \overline{z}\,\mathrm{d}z.$$

Suggestion. Note that expression (4), Sec. 50, can be used here even though the function $f(z) = \overline{z}$ is not analytic anywhere [see Example 2, Sec. 19].

Solution. Write $f(z) = \overline{z} = x - iy = u(x, y) + iv(x, y)$, so $u = x$ and $v = -y$. Expression (4) of Sec. 50 follows from Green's theorem and remains valid for this continuously differentiable, nonanalytic function:

$$\int_C f(z)\,\mathrm{d}z = \iint_D (-v_x - u_y)\,\mathrm{d}A + i\iint_D (u_x - v_y)\,\mathrm{d}A,$$

where D is the region enclosed by C. Since

$$-v_x - u_y = 0, \qquad u_x - v_y = 1 - (-1) = 2,$$

we get

$$\int_C \overline{z}\,\mathrm{d}z = 2i\iint_D \mathrm{d}A = 2i\,\mathrm{Area}(D).$$

Thus

$$\boxed{\mathrm{Area}(D) = \frac{1}{2i}\int_C \overline{z}\,\mathrm{d}z.}$$

□